

Annotation Guidelines for the Linguistic Quality Metric (LQM)

The Linguistic Quality Metric (LQM) is a fine-grained evaluation framework developed to support detailed human assessment of machine translation output for Arabic dialects. LQM builds on the general principles of existing quality evaluation frameworks, such as MQM, while introducing linguistically motivated categories and subcategories that better capture phenomena specific to Arabic dialectal variation and usage.

LQM was developed through iterative qualitative analysis of human evaluations of machine translation outputs across multiple Arabic dialects. During this process, recurring linguistic error patterns were identified that are not adequately distinguished in existing evaluation schemes. LQM addresses this gap by organizing errors into **six high-level categories** based on the primary linguistic dimension affected, including:

1. Sociolinguistic,
2. Pragmatic,
3. Semantic,
4. Morphosyntactic,
5. Orthographic/ Writing conventions and,
6. Graphetic.

These guidelines define the LQM categories and provide instructions to support consistent and replicable annotation:

1. Sociolinguistics: Code & Register Selection

This category captures errors involving inappropriate or inconsistent use of language varieties, registers, or socially meaningful linguistic choices. Sociolinguistic errors occur when the translation fails to respect the intended dialect, level of formality, or social norms of the target language, relative to the task specifications, the source text, or established usage conventions. Such errors do not necessarily involve incorrect meaning or grammatical structure, but rather reflect a mismatch in variety choice, register, or social meaning, including misuse or mixing of dialects and inappropriate register selection.

1.1. Standardization Interference (Vertical Mismatch)

A Standardization Interference error occurs when the model uses a standardized, prestige, or “high” variety of a language instead of the specified target vernacular or dialectal variety. This typically reflects interference from a standard language (e.g., MSA, Standard German, Standard English) in contexts where a vernacular or “low” variety form is required.

In Arabic dialect translation, this error commonly manifests as the use of Modern Standard Arabic (MSA) vocabulary, morphology, or syntax in contexts where a regional dialect is expected. Although MSA may be grammatically correct and intelligible, its use alters the sociolinguistic meaning and violates the intended variety.

Example:

English (Source): I swear my clothes are light. May Allah guide you. It's hot. Turn on the air conditioner.	UAE Arabic (Target): أقسم لك ملابسي خفيفة. اللهم اهده. حار . شغل مكيف الهواء
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The translation consistently uses **MSA lexical and syntactic construction** forms rather than Emirati or Gulf dialectal equivalents. Indicators of standardization interference include **أقسم لك، مكيف الهواء**

1.2. Wrong Dialect (Horizontal Mismatch)

A Wrong Dialect error occurs when the translation contains lexical items, idioms, particles, or constructions characteristic of **another dialect** other than the specified one. This represents a **horizontal mismatch** across dialects at the same sociolinguistic level.

Example:

English (Source): Do you still remember? Oh man, the airports and bridges are closed, how can we travel? Oh man, we sold the lamp to buy tobacco!	Palestinian Arabic (Target): إنتا لسه بتذكر؟ يا ربي ، بيغنا المطارات والجسور مغلقة، كيف رح نسافر؟ يا ربي، بيعنا عشان نشترى تبغ!
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The translation includes features associated with other Levantine or non-Palestinian varieties: Lexical choices like **عشان** (commonly associated with Egyptian and some Levantine usage) instead of more Palestinian-preferred alternatives.

1.3. Register Mismatch

Characteristics of text that uses a level of formality higher or lower than required by the specifications or general language conventions.

Example

Jordanian (Source): يا بيبك باختصار يعني و بدون لف و دوران، أنا بطلع لي أوفر تايم	English (Target): Look, dude , to be brief and without beating around the bush, I'm going to get some overtime.
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The English translation is more formal than the casual, colloquial tone of the original Arabic

2. Pragmatic: Use, Context, Cultural Appropriateness

Errors in this category occur when the language choices in a text fail to align with pragmatic norms, including the intended use, context, or cultural expectations. These errors can affect how meaning is interpreted, how politeness or intent is conveyed, or whether the text is socially or culturally appropriate.

2.1. MWEs/ Proverbs

This error type refers to cases where the model fails to correctly identify or render multi-word expressions, idioms, or proverbs and instead produces a literal, non-idiomatic, or incorrect formulation. Such errors arise when the meaning of a fixed or formulaic expression in the source language is not preserved through an appropriate target-language equivalent, or when an idiomatic expression is misused in context. As a result, the output may sound unnatural, culturally inappropriate, or may distort the intended meaning despite being grammatically well-formed.

Example

Egyptian (Source): و نعم بالله طب ربحيني مغيث حد كده و لا كده بينغيش في قلبك؟	English (Target): And yes by God, don't you have anyone like that , who doesn't make you feel bad?
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The expression “حد كده ولا كده” is a colloquial Egyptian Arabic construction that functions pragmatically rather than referentially. It does not mean “anyone like that” in a concrete or descriptive sense; instead, it signals indeterminacy and speaker alignment, roughly conveying “someone, anyone, one way or another” or “someone who more or less fits the bill.” Translating it literally as “anyone like that” misses this pragmatic nuance and makes the utterance sound more categorical and direct than intended.

Example

Egyptian (Source): لا ساخت ايه ده حتى دمها تقيل و دمها يلطش منتيش شايقة دي عاملة زي الفازة المقطوشة	English (Target): Al didn't do anything, her blood is thick and sticky , you don't see? She's like a cut vase.
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In Egyptian Arabic, “دمها تقيل” is a fixed idiomatic expression describing someone as unpleasant, socially awkward, boring, or lacking charm. It has no relation to physical blood properties. Rendering it as “her blood is thick” constitutes a clear failure to recognize the idiom, producing a phrase that is meaningless and confusing in English.

The follow-up expression “دمها يلطش” intensifies the negative evaluation, implying that the person's presence is irritating or emotionally draining. Translating it as “sticky” again reflects literal lexical mapping rather than idiomatic meaning.

2.2. Code Switching

This error type refers to failure of handling foreign words, hence, implying a failure to recognize a shift in linguistic code.

Example

English (Source): Mohieddin, my brother, I want you to bring me twelve bananas , yes, and a box of tangerines..	UAE Arabic (Target): حي الدين، خوي، أبليك تجيب لي ١٢ بنانة ، إيه، وعلبة يوسف
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In UAE Arabic, the translation fails to maintain a consistent linguistic code, where a foreign word is introduced where a natural Arabic equivalent exists (موز). Hence, the code switch is unmotivated and unnecessary as the communicative situation assumes a monolingual Arabic audience.

2.3. Speech Acts/ Illocutionary Force

This error occurs when the intended meaning (illocutionary force) of an utterance in the source language is not accurately conveyed in the target language due to cultural or linguistic differences/when the translator fails to convey the speaker's intention (like promising, ordering, or questioning) behind the words, instead just translating the literal meaning

Example

Yemeni Arabic (Source): اين جنة، انا ما اشتيتش بالجنة اشتهي يعوني بالدنيا اشتهي حجي ، اشتهي ذهبي ، بالرضى بالصميل ، اشتهي ذهبي	English (Target): Where is the paradise? I don't want paradise; I want my rights in this world. I want my pilgrimage, I want my gold, by consent or by force, I want my gold.
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In this example, the phrase اين جنة is literally translated into "Where is paradise?" However, in this context, the speaker is not genuinely asking for information about the afterlife. Instead, it functions as a Rhetorical question, expressing frustration or rejection of the promised reward in the afterlife. A more pragmatic rendering would be "What good is paradise to me?"

2.4. Discourse Marker Mismatch

This error occurs when there is an inconsistency between the semantic/pragmatic meaning of a discourse marker and the context in which it is used. This can lead to a lack of coherence or an unintended meaning in a conversation or text.

Example

Egyptian Arabic (Source): والله طلبك هو الي هيحدد إذا كنت أقبليه ولا أرفضه	English (Target): God willing , your request is what will determine whether I accept or reject it.
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In Egyptian Arabic, “والله” is a highly context-dependent discourse marker. While its literal meaning is “by God,” in everyday speech it rarely functions as a literal oath. Instead, it often serves to emphasize the truth or reality of a statement, roughly equivalent to “actually,” “in fact,” or “truly” in English. In the source text, the function of والله is not to express hope or contingency but to assert the fact that the request determines the outcome. Translating it as “God willing” misrepresents this function because “God willing” conveys tentativeness or futurity, whereas the original Arabic emphasizes certainty and factual reality. Pragmatically, a closer English rendering would be “Actually, your request is what will determine whether I accept or reject it” or “In fact, your request decides this.”

2.5. Vocatives /Honorifics/Titles

This error occurs when the translator fails to preserve the pragmatic and social functions of vocatives, honorifics, or titles in the source language. Vocatives are forms of direct address (e.g., names or terms used to call or refer to someone in speech), while honorifics and titles indicate respect, social status, professional rank, or politeness. The error arises when these elements are omitted, mistranslated literally, or rendered inappropriately, resulting in a loss of social nuance or altered tone.

Example

Egyptian Arabic (Source): رفقي بيه عزيز يبقى اللي فاضل من المطلوب التحقيق معاهم فوزي الدالي هو في اسكندرية ان شاء الله هيوصل آخر النهار النهاردة.	English (Target): Refki with Aziz , so the rest of those who are wanted for investigation are Fouzzy Dali, who is in Alexandria, God willing, will arrive by the end of the day today.
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In Egyptian Arabic, the title بيه carries respectful, vocative force. Translating it incorrectly (with) erases politeness and clarity, creating a mistranslation of forms of address. Proper rendering should preserve honorific function.

3. Semantics

3.1. Error type: Lexical Semantics

Errors in this category occur when the choice, meaning, or expression of a word or phrase in the translation does not accurately reflect the source text, causing semantic distortion, ambiguity, or unnatural expression.

3.1.1. Named Entity (NE) – Reference Resolution Failure

This error occurs when the translation fails to correctly map a proper noun in the source text to its referent in the target text. This includes mistranslating, omitting, or generalizing the named entity, including cases where pronunciation differs from standard orthography.

Example

UAE Arabic (Source): يمعة ، أي شي أمي تقوله و لا يدو، و لا أبوي، أهو إلي يمشي، هذيلا هم الخير و البركة	English (Target): Honestly , whatever my mother says goes, not my father, it's her way that works, they are the good and the blessing
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In this example, the proper noun **يمعة (Jumaa)** in UAE Arabic is replaced with a generic term “a mess.” Reference resolution fails, and the entity's identity is not conveyed.

3.1.2. Wrong Term

Use of a term that, while it may convey the general semantic meaning, is not the term a domain expert would use, or gives rise to a conceptual mismatch in the professional, technical, or domain-specific context.

Example

English (Source): This is the honor of the profession. My client asked for help from me, so I must support her.	Egyptian Arabic (Target): عميلتي ، دي مهنتي، طلبت مني المساعدة، و أنا لازم أساعدها.
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In this example, the term **عميلتي** correctly conveys “my client” instead of the domain-specific **موكلتي**, which is appropriate in legal/professional contexts. This creates a conceptual mismatch, because the source emphasizes professional duty and ethics, not just a casual client relationship.

3.1.3. Overtranslation

Using a more specific term (hyponym) in the translation when the source text uses a general term (hypernym), adding information that was not present in the source. This can introduce unintended meaning or mislead the reader.

Example

Moroccan Arabic (Source): أنا لي هبيلة. كنقول لهم النهار: دالعرس دبنتي شامة أنا غادي نوقف فيه بحال لي واقفة مها الله يرحمها	English (Target): I have a disability . I tell them that today is the wedding of my daughter Chama, and I'm going to stand there like I'm standing with her, God rest her soul
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The source term هبيلة is a general self-reference that may imply “foolish, silly, or mentally impaired” in a colloquial sense, without necessarily indicating a formal disability. Translating it as “I have a disability” introduces a more specific medical or legal category (hyponym) that is not in the source, creating an overtranslation. This could mislead readers into thinking the speaker has a formal physical or cognitive disability, which is unintended.

3.1.4. Undertranslation

Using a more general term in the target when the source uses a specific term, resulting in loss of important information. The translation conveys less detail than the source, which may reduce clarity or neglect nuances intended for the reader.

Example

UAE Arabic (Source): يا يدو فكينا من البراقع، نحن في: الألفية الثالثة و هي بعدها تطرلنا براقع، لبسي نقاب و فكي عمرج	English (Target): Hey, take off the veils , we're in the 21st century, and you're still wearing veils, wear a niqab and take off your niqab.
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In the example, the source term براقع refers specifically to burqas (full-body covering), but it is translated as “veils”, a more general term. This undertranslation loses the specificity of the source, which contrasts different types of face/body coverings (burqa vs niqab). The nuance and cultural specificity are not preserved.

3.1.5. Transliteration

Providing a transliteration of certain words that are phonetically rendered into the target language incorrectly.

Example

English (Source): Can you tell your father that I will come to visit you at home and have helba with him?	Egyptian Arabic (Target): هتقولي لأبوك هاجي أزورك في البيت وأكل هلبة معاه؟
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The source word “helba” is a transliteration of a culturally specific food item. While the meaning is generally preserved, the Arabic هلبة may not accurately reflect the intended pronunciation or the

conventional rendering of this dish in Egyptian Arabic. This constitutes a transliteration error, because phonetic fidelity and standard naming conventions are not fully respected.

3.1.6. Unnatural / Unidiomatic Style

This error occurs when the translation is grammatically correct, but the wording or phrasing sounds awkward, non-native, or unnatural in the target language. This reduces readability and fluency even though no grammatical rules are violated.

Example

Egyptian Arabic (Source): ولاد الدلي بعنك فنون جميلة عشان تسجل المواجهة ترجعلي بحدوتة مع بنت سعد الدالي؟	English (Target): the sons of the deli? I sent you to the Faculty of Fine Arts so you can record the confrontation, and you come back with a story with a girl from Saad El Dali?
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The translation is grammatically correct but awkward and unnatural. The phrasing “a girl from Saad El Dali” in English suggests the girl is geographically “from” Saad El Dali, rather than being his daughter. A more idiomatic rendering would be “Saad El Dali’s daughter” or “with a story involving Saad El Dali’s daughter.” The literalness disrupts fluency and fails to convey the intended kinship meaning.

3.1.7. Awkward Style

Style involving excessive wordiness or overly embedded clauses, often due to inappropriate retention of source text style in the target text.

Example

Egyptian Arabic (Source): و القرار ده حيتنفذ حيتنفذ دي مسألة بيتخطط لها مزبوط و بتتدرس من زمان قوي اسمع كلامي بقى أنا جيلك و معايا الحل.	English (Target): and this decision will be implemented, it will be implemented , this is a matter that has been planned properly and has been studied for a long time, listen to me now, I'll get it to you, and I have the solution.
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The repetition ‘this decision will be implemented, it will be implemented’ mirrors Arabic rhetorical emphasis but results in awkward redundancy in English, where such repetition is stylistically unnatural.

3.1.8. Coverage: Unknown Term / Dialect

This error occurs when the model fails to recognize a specific dialectal lemma

Example

Egyptian Arabic (Source): هه بص الخيابة اللي فيها الواد قال بتعاكسك؟	English (Target): Hey, look at the audacity of this guy, he said he was hitting on you?
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The Egyptian Arabic word **خيابة** is a dialect-specific lemma meaning naivety or foolish innocence, typically used sarcastically to mock someone's lack of awareness. In the translation, it is rendered as "audacity", which reflects an opposite evaluative meaning and indicates that the model failed to recognize the dialectal lemma altogether. Rather than identifying **خيابة** as a term expressing gullibility, the model inferred a generic negative trait, resulting in semantic distortion.

3.1.9. Disambiguation: Polysemy Failure

The model knows the word but picks the wrong meaning (your "Sign" vs "Knock down" example).

Example

Moroccan Arabic (Source): إيوا تا نتي ما لقاكش حارة . كن كنت أنا فيلاصتك كن والله نا نوريه خيزو ما حلاه	English (Target): Okay, you didn't find warmth either. If I were in your place, I swear I'd show him who he's dealing with and wouldn't let him get away with it.
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The Moroccan Arabic adjective **حارة** is polysemous. While one sense refers to physical heat or warmth, in this context it is used to mean intensity, toughness, or harsh treatment (i.e. someone didn't go hard on him / didn't respond firmly). The model correctly recognizes the word but selects the literal thermal sense ("warmth"), which is pragmatically inappropriate in the given context.

3.1.10. Disambiguation: Cross-Variety Interference

The model knows the word but assigns it the Standard meaning instead of the Dialect meaning.

Example

English (Source): All right, dad, make yourself comfortable, sit with your friend. I have some work, so I will be off	UAE Arabic (Target): خلاص يوبا، ارتاح، اقعد ويا ربيّك. عندي شوية شغل، باخذ إجازة .
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The model translates 'I will be off' as **إجازة**, applying the Standard Arabic meaning ('formal leave/holiday') instead of the UAE dialectal nuance ('temporary break/stepping aside'), resulting in a cross-variety disambiguation error.

3.1.11. Unintelligible

This error occurs when the translation output is garbled, ungrammatical, or incomprehensible, such that a reader cannot recover the intended meaning from the target text.

Example

English (Source): Your request is what will determine whether I accept it or reject it. .	Egyptian Arabic (Target): طلبك هو اللي هيديكّر قبولي أو رفضي ليه
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The word 'هيديكّر' is garbled and incomprehensible; it does not exist in Arabic and prevents understanding of the sentence, resulting in an Unintelligible error.

3.1.12.Measurement Units

"Inappropriate measurement format for its locale.

Examples:

- A text created for use in France uses feet and inches and Fahrenheit temperatures."
- Decimal Separator (Comma vs. Period): Arabic often uses comma (,) as a decimal separator, while English uses period (.): Arabic: ٣,٥ Wrong: 3,5 Correct: 3.5
- Unit Formatting (e.g., % Signs, Currency): Arabic places % signs and currency in different positions or formats: Arabic: %٥ Incorrect: %5 Correct: 5%"

3.2. Error type: Propositional Semantics

This error type concerns the meaning of propositions (entire statements or clauses) rather than individual words. It occurs when the translation adds, omits, leaves untranslated, or invents content, affecting the truth conditions or overall information conveyed by the source text.

3.2.1. Addition

Error occurs when the translation includes information not in the source text.

Example

Mauritanian Arabic (Source): مانى ليهي نحصر ان شاء الله، الا كايسة الحمام و لين نوف منو جايتكم ان شاء الله، ماني ليهي نحصر وفي ذاك.	English (Target): I'm not going to be brief, God willing, except for the pigeon's call and the new moon's arrival , God willing, I'm not going to be brief in that
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The translation introduces 'new moon's arrival,' which is not in the source, resulting in an Addition error.

3.2.2. Omission

This error occurs when information present in the source is missing in the translation, resulting in loss of propositional content. The omitted information can affect meaning, nuance, or context.

Example

Palestinian Arabic (Source): قبل، بعد، ما بفهم أنا. المهم النية بكت موجودة، معناها إنته مكورن معهم يعني مكورن معهم. و مش لحالك، أنا بعرف كل القصة، إنته مكورن و مسعود الحذر معكم	English (Target): Before, after... I don't get it. The important thing is the intention was there, which means you were in on it with them—I mean, you were in on it with them. And not by yourself; I know the whole story. You were in on it, and Mas'oud al-Hathar was with you all.
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The translation omits the proposition 'مكورن' (infected with corona), replacing it with unrelated content, resulting in a loss of critical information.

3.2.3. Untranslated

This error occurs when a segment intended for translation remains in the source language and is carried over into the target text.

Example

UAE Arabic (Source): نحن عاذرينك يا راجح، و أي شي في مصلحة البلاد نحن بنتعاون وياك و بنحط إيدينا فايديك، و لا شو رايك يا عبيد ؟	English (Target): We forgive you, Rajeh, and anything in the country's interest, we cooperate with you and put our hands in your hands, and what do you think, عبيد ?
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The model leaves the proper noun 'عبيد' untranslated in the target English sentence, reducing comprehensibility and resulting in an Untranslated error.

3.2.4. Hallucination

A hallucination occurs when the translation introduces new information - adjectives, nouns, actions, or other propositions - that is not present in the source text, changing the factual content or meaning. Unlike Addition, hallucination often fabricates details that are specific, concrete, or assertive, potentially misleading the reader.

Example

English (Source): Calm down, calm down, sir. If the issue isn't worth it, would I have come here in this heat and left my bride?	Egyptian Arabic (Target): إهدى يا بيه، إهدى. لو الموضوع ميكسرش إزازه، كنت قعدت في العز و سيبت عروسة؟
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The translation introduces the phrase 'ميكسرش إزازه' ('doesn't break a glass'), which is not present in the English source. This adds a concrete metaphorical detail that changes the factual content of the proposition, the source simply evaluates the importance of the issue ('if the issue isn't worth it').

3.3. Error type: Discourse Semantics

Discourse Semantics errors occur when meaning relationships, coherence, or consistency across sentences or larger units of text are disrupted. Unlike lexical or propositional errors, these affect how information is structured, tracked, and understood across a text.

3.3.1. Inconsistent Use of Terminology

This Occurs when multiple terms are used for the same concept within a text in situations where consistency is desirable, which can confuse readers or reduce clarity.

Examples:

The text refers to a component as the 'brake release lever', 'brake disengagement lever' , 'manual brake release', and 'manual disengagement release'.

3.3.2. Inconsistent with terminology resource

Use of a term that differs from term usage required by a specified termbase or other resource.

Examples:

- 1) A termbase specifies that the term 'USB memory stick' should be used, but the text uses 'USB flash drive'.
- 2) A French text translates English 'e-mail' as 'e-mail' but terminology guidelines mandated that 'courriel' be used.
- 3) The English musicological term 'dog' is translated (literally) into German as 'Hund' instead of as 'Schnarre', as specified in a terminology database.

3.3.3. Pronouns

Failure to correctly render pronouns, causing changes in speaker, referent, or flipping the intended gender, which can mislead the reader or alter the discourse meaning.

Example

Moroccan Arabic (Source): واش انا عارفاتي ميرزط ما عارف شكون يفرش ليا البيرو و البيت نزيد نكملها ب باباك الله يخلف عليه باراكا عليه صايب لينا لايباج	English (Target): Am I not aware that I'm stressed? I don't know who will furnish my office and the room. And on top of that, your father—may God bless him—he's had enough, he should make this whole elaborate setup for us.
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The phrase 'Am I not aware that I'm stressed?' misrenders the implicit second-person reference in 'واش انا', changing the speaker-listener dynamic and the intended discourse meaning.

3.3.4. Inconsistent Style

Style that varies inconsistently through the text, such as switching between formal and informal, technical and colloquial, or literary and plain styles, which disrupts coherence.

Examples:

- 1) One part of a text is written in a clear, "terse" style, while other sections are written in a more wordy style.
- 2) The same text recurs at several points in a large document that has been divided up and submitted to multiple translators, with the result that that text is translated in three different ways, which can involve different style as well as terminology or register differences.

4. Morphosyntactic

4.1. Error type: Grammar, wrong number, gender, verb tense

Morphosyntactic errors occur when the translation violates the grammatical rules of the target language, affecting the morphology (form of words) or syntax (how words are combined in sentences). These errors disrupt agreement, tense, number, gender, case, definiteness, or verb features, and can make the sentence ungrammatical or confusing.

4.1.1. Verbal Features

Error that occurs when a text string (sentence, phrase, other) in the translation violates the grammatical rules of the target language./ Tense, Aspect, Voice, Mood, Person.

Example

English (Source): My mother is very kind, and we should be happy that they reconciled , not asking why they reconciled .	Egyptian Arabic (Target): أمي طيبة أوي، و المفروض نكون فرحانين إنهم صالحوا ، مش نسأل ليه صالحوا
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The error concerns verb valency and transitivity. The English verb “reconcile” is rendered as **صالحوا** in the target, which is treated as a transitive verb without an object. In Arabic, **صَالَحَ** is obligatorily transitive and requires an explicit object or prepositional complement (e.g. **صالحوا بعضهم / صالحوا فلان**). Using **صالحوا** in isolation violates Arabic verbal subcategorization rules, resulting in an ungrammatical structure. In this context, the correct Arabic equivalent should be the intransitive reciprocal form **اتصالحو** or a transitive form with an explicit complement. The error is therefore a Verbal Features error involving voice and argument structure, where a transitive verb is incorrectly used as if it were intransitive, producing a morphosyntactic violation in the target language.

4.1.2. Nominal Features

A Nominal Features error occurs when the translation violates the grammatical rules governing nouns and noun phrases in the target language. Unlike verbal features (which concern verbs), this category focuses on how nouns are morphologically and syntactically marked. Such errors often sound immediately “wrong” to native speakers, even if the overall meaning is still recoverable. They concern (Number, Gender, Case, Definiteness, State [Idafa])

Example

Mauritanian Arabic (Source): يا خياتي راني لكم فاصل في نعرف شي لراسي هذا ما هو معقول بذي الصفة	English (Target): Oh, my brothers , I’m taking a break, I need to clear my head, this is not reasonable with this situation.
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The error in this example involves a gender mismatch in the translation of **خياتي**, which is a feminine plural address form in Mauritanian Arabic meaning “my sisters.” Rendering it as “my brothers” incorrectly flips the grammatical gender of the noun. This is a Nominal Features (Gender) error because the translation violates the gender specification encoded in the source noun, altering the intended addressee group and potentially misleading the reader. Even though the discourse function (a form of address) is preserved, the gender feature is incorrectly realized, which is grammatically and pragmatically significant in Arabic.

4.2. Error type: constituent order

4.2.1. Address Format:

Error involving inappropriate address format for locale.

Examples:

<u>1. Order Reversal</u> Arabic source: الرياض، حي النرجس، شارع الأمير محمد بن سلمان Incorrect MT: Riyadh, Al Narjis, Prince Mohammed bin Salman Street (Still in Arabic order; lacks proper abbreviations.) Correct (English style): Prince Mohammed bin Salman St., Al Narjis District, Riyadh	<u>2. Missing Abbreviations & Prepositions</u> Arabic source: بيروت، شارع الحمرا، لبنان Incorrect MT: Beirut, Hamra Street, Lebanon (Acceptable but not fully formatted for international mail.) Improved: Hamra Street, Beirut, Lebanon, P.O. Box 12345
<u>3. Arabic Numerals Left Unconverted</u> Arabic source: المنصورة، شقة ٢٣، الدور الثالث، عمارة رقم ١٥ Incorrect MT: Mansoura, 3 الدور, 23 شقة, Building No. ١٥ (Contains untranslated Arabic terms and numerals.) Correct: Apt. 23, 3rd Floor, Building No. 15, Mansoura"	

4.2.2. Date format

Error involving inappropriate date format for its locale.

Examples:

<u>Date Format Differences:</u> Arabic follows day/month/year; English may prefer month/day/year depending on locale. Arabic: ٢٣/١٠/٢٠٢٢ Wrong (direct): 23/10/2022 (if targeting U.S. English) Correct (U.S.): 10/23/2022"

5. Orthographic/ Writing conventions

5.1.Spelling

Errors involving the incorrect written form of words according to standard or accepted conventions in the target language.

Example

English (Source): You want to know why I doubt Khaled El-Dali specifically, don't you?	Egyptian Arabic (Target): عايز تعرف ليه أنا شاك ف خالد الدالي بالذات، مش كده؟
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The target translation “شاك ف خالد” contains three spelling errors. First, شاك should be شاكك to correctly convey the intended meaning of doubting, as the missing ك changes the meaning and grammatical form. Second, the preposition ف should be spelled في to follow standard written Egyptian Arabic conventions. Third, خالد contains an extra هاء at the end. While the sentence remains understandable, these misspellings violate orthographic norms and reduce both clarity and accuracy. The correct version is: شاكك في خالد الدالي بالذات.

5.1.1. Inconsistent Spelling

The same word is spelled in different ways within the same text. It often occurs with proper nouns, loanwords, or transliterated items.

5.1.2. Unconventional Spelling

A spelling that is non-standard, informal, or idiosyncratic when a conventional form exists. It is Common in dialect-to-standard or speech-to-text translations. It makes the word hard to read, even if technically phonetic.

5.2. Surface Mechanics

Surface Mechanics errors involve violations of basic writing conventions that affect the visual form and readability of the text rather than its meaning. While such errors usually do not change propositional content, they disrupt reading flow and signal poor editorial quality or lack of fluency in the target language.

5.2.1. Number Format

Inappropriate number format for its locale.

Example

English (Source): They accused me for some crimes. If they tried me for them, I would not get out from them before two thousand and ninety-nine .	Egyptian Arabic (Target): اتهموني في بعض الجرائم. لو حاكموني فيهم، مش هخرج منهم قبل ألفين وتسعة وتسعين .
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In the target translation, writing ألفين وتسعة وتسعين incorrectly adds a “و” before the units (“تسعة”), which is not used in Egyptian Arabic. The correct form is: ألفين تسعة وتسعين.

Other Example:

1. Digit Directionality / Reversal: _Mixing right-to-left digit order (from Arabic) can lead to reversed or jumbled numbers. Wrong: 005,21 instead of 12,500 Correct: 12,500	2. Arabic vs. Western Numerals (١٢٣٤ vs. 1234): _MT may incorrectly copy over Arabic-Indic numerals, which should be converted to Western numerals in English. Wrong: The price is ١٠٠ pounds. Correct: The price is 100 pounds.
3. Incorrect Conversion of Ranges and Approximations: Arabic: بين ١٠٠٠ و ٢٠٠٠ شخص Wrong: Between 1000 and to 2000 people Correct: Between 1,000 and 2,000 people	

5.2.2. Currency

Incorrect currency format for its locale.

Example

English (Source): The fellow will give me ten rubies , father. I told you I went and talked to him, and now she has agreed.	UAE Arabic (Target): الرّيال بيعطيني عشر ياقوتات ، يوبا. قلتلك رحت و كلمته، و الحين هي وافقت
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In this example, the target treats *rubies* as gemstones rather than recognizing it as a currency unit in the source context. This results in an incorrect currency rendering for the target locale, where monetary amounts are conventionally expressed using a recognized currency name (e.g., روبيات).

5.2.3. Time format

Error involving incorrect time format for its locale.

Examples

1. 12-Hour vs. 24-Hour Format _Arabic source: الساعة ٣ بعد الظهر Incorrect MT: 3 PM (While understandable, this mixes with other styles and lacks format consistency.) Recommended (for formal/technical contexts): 15:00 (24-hour format)	2. Missing Leading Zeros _Arabic source: ٠٧:٠٥ صباحًا Incorrect MT: 7:5 AM (Missing zero padding reduces clarity.) Correct (ISO style): 07:05
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3. Misplaced AM/PM Indicator Arabic source: الساعة 12:30 منتصف الليل Incorrect MT: 12:30 PM (Mixes midnight with “PM”, completely reversing meaning.) Correct: 12:30 AM	4. Dropping the “T” or Time-Zone in ISO Format Arabic source: ٠٠٠٠+١٦:٤٥:٠٠ ١٧-٠٦-٢٠٢٥ Incorrect MT: 2025-06-17 16:45:00 (Missing ""T"" delimiter and timezone.) Correct: 2025-06-17T16:45:00Z
5. Arabic-Indic Numerals Left Unchanged Arabic source: الساعة ١٤:٣٠ Incorrect MT: ١٤:٣٠ (Digits remain in Arabic-Indic, unreadable to Western readers.) Correct: 14:30"	

5.2.4. Telephone

Error involving inappropriate telephone number form for locale

Examples:

1. Missing Country Code / Incorrect Prefix Arabic Source: اتصل بي على ٠٥٣١٢٣٤٥٦٧ Incorrect MT: Call me at 0531234567 (Local format only—misleading for international users) Correct (E.164 international format): +966 531 234 567 ! Saudi Arabia's country code is +966, and for international calls, the leading “0” must be dropped	2. Wrong Area Code Handling Arabic Source: اتصلوا بمكاتبنا في بيروت على ٠١٢٣٤٥٦ Incorrect MT: Call our Beirut office at 0123456 (Kept local prefix “0” but no country code) Correct (International): +961 1 234 56 → Lebanon: country code +961, drop the leading “0” from area code “01” .
3. Misgrouped Digits / Lack of Formatting Arabic Source: رقم الطوارئ في السعودية هو ٩٩٩ Incorrect MT: The emergency number in Saudi Arabia is 999 (This one is correct as a local number, but grouping should be maintained for clarity.) Best Practice (Formatted Even Locally): 999 is fine, but critical numbers still benefit from clarity: Emergency (Saudi): 999	4. Use of Arabic-Indic Numerals in English Output Arabic Source: اتصلوا على ٩٦٦٥٣١٢٣٤٥٦٧+ Incorrect MT: Contact us at +٩٦٦٥٣١٢٣٤٥٦٧ (Digits left in Arabic-Indic—English readers might misread or ignore them.) Correct: +966 531 234 567 → Western digits and space-separated format ensure readability ".

5.3.Punctuation

Punctuation incorrect according to target language conventions. Even though punctuation errors may seem minor, they can significantly impact the readability and clarity of a sentence. It is crucial to examine

the meaning in Arabic and assess how the model handled it in translation, ensuring that the intended meaning is accurately conveyed in the target language. Translators should check punctuation to reflect natural sentence boundaries, disambiguate meaning when needed, and ensure coherence, especially in lists, dialogues, and compound sentences, without copying source-language punctuation patterns blindly.

Example

English (Source): It would not be appropriate. I told you, I have a driver from the group with me all the time.	Egyptian Arabic (Target): مش مناسب، قلتك، معايا سائق من المجموعة دائماً
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In the target translation, the commas in are used incorrectly according to Arabic punctuation conventions. The placement of the second comma after “قلتك” unnaturally fragments the sentence, interrupting the natural reading rhythm and making the flow feel choppy.

6. Graphetic

6.1.Character Encoding

Error occurring when characters garbled due to incorrect application of an encoding. This type of error can arise from machine translation when unusual characters appear in the output, including characters from languages other than the target language. These errors are generally easy to detect and should be labeled as character encoding errors.

Example

Jordanian Arabic (Source): فواز صدقتي أنا ما عرفت إلا اليوم الصبح و ما حبيت أدايقك، و كنت منتظر الوقت المناسب عشان أحكيك في الموضوع	English (Target): Fواز believe me, I only found out this morning and I didn't want to bother you, and I was waiting for the right time to tell you about it.
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In the target translation, the first word “Fواز” is a clear character encoding error. The Latin letter “F” has been incorrectly inserted before the Arabic word “فواز”, likely due to a mismatch in character encoding or a parsing issue in the machine translation output. This results in garbled text that mixes scripts inappropriately, which can confuse the reader and disrupt readability.

The rest of the sentence is correctly translated, but the presence of the corrupted word violates standard encoding rules.

LQM MT Annotation Guidelines — Summary Table

This table summarizes the LQM MT annotation guideline categories, error types, subcategories, short descriptions, and illustrative examples.

Category	Error type	Subcategory	Short description	Example
Sociolinguistics	Code & Register Selection	Standardization Interference (Vertical mismatch)	Uses a standard / prestige variety (e.g., MSA) instead of the specified dialect/vernacular, changing sociolinguistic meaning even if grammatical.	EN: "It's hot. Turn on the air conditioner." → UAE AR uses MSA-like forms: حار، ملابسي، أقسم لك، مكيف الهواء rather than Emirati/Gulf equivalents.
Sociolinguistics	Code & Register Selection	Wrong Dialect (Horizontal mismatch)	Uses features from the wrong dialect (lexicon/particles/idioms/constructions not characteristic of the target variety).	EN → Palestinian AR includes عشان (flagged as associated with Egyptian / some Levantine usage), violating target variety preference.
Sociolinguistics	Code & Register Selection	Register Mismatch	Output is more/less formal than required by spec or by the source's tone/conventions.	Jordanian source is colloquial; English target becomes noticeably more formal than the casual tone.
Pragmatic	Use, Context, Cultural Appropriateness	MWEs / Proverbs	Fails to recognize/render idioms/MWEs/proverbs, producing literal/unidiomatic meaning (often grammatical but pragmatically wrong).	Egyptian: "حد كده ولا كده" translated literally as "anyone like that" → misses pragmatic function ("someone, anyone, one way or another / fits the bill").
Pragmatic	Use, Context, Cultural Appropriateness	MWEs / Proverbs	Fails to recognize/render idioms/MWEs/proverbs, producing literal/unidiomatic meaning (often grammatical but pragmatically wrong).	Egyptian: "دمها ثقيل / دمه" rendered as "her blood is thick... sticky" → literal mapping, idiom not preserved.
Pragmatic:	Use, Context, Cultural Appropriateness	Code Switching	Unmotivated/incorrect foreign word insertion or failure to handle foreign words (code shifts) appropriately.	EN "bananas" → UAE AR uses بانه where a natural Arabic equivalent موز exists; switch is unnecessary for monolingual Arabic audience.
Pragmatic	Use, Context, Cultural Appropriateness	Speech Acts / Illocutionary Force	Literal translation fails to convey speaker intent (request/promise/order/rhetorical question, etc.), yielding pragmatic misfire.	Yemeni: "اين الجنة" translated "Where is paradise?" but functions rhetorically; closer: "What good is paradise to me?"

Pragmatic	Use, Context, Cultural Appropriateness	Discourse Marker Mismatch	Discourse marker doesn't match context; harms coherence or intended nuance.	Egyptian والله translated "God willing" (tentative) though source uses it as emphasis "actually / in fact / truly."
Pragmatic	Use, Context, Cultural Appropriateness	Vocatives / Honorifics / Titles	Fails to preserve social/pragmatic function of address forms, honorifics, titles (omit, mistranslate, overly literal).	Egyptian "بيه" (respectful title) is mistranslated as "with," losing politeness/clarity.
Semantics	Lexical Semantics	Named Entity (NE) – Reference resolution failure	Proper noun is mistranslated/omitted/generalized; entity identity not conveyed.	UAE AR جمعة (Jumaa) rendered as "a mess" (generic), losing referent.
Semantics	Lexical Semantics	Wrong Term	Uses non-expert/non-domain term that creates conceptual mismatch in a technical/professional context.	EN "client" in legal sense → Egyptian uses عميلتي instead of domain-appropriate موكلتي.
Semantics	Lexical Semantics	Overtranslation	Target is more specific than source (hyponym for hypernym), adding unintended detail.	Moroccan هيبلة → "I have a disability" (too specific/medicalized).
Semantics	Lexical Semantics	Undertranslation	Target is more general than source, losing key specificity/nuance.	UAE برقع rendered "veils" (general) instead of specific covering; loses contrast (burqa vs niqab).
Semantics	Lexical Semantics	Transliteration	Uses transliteration incorrectly / non-conventional phonetic rendering.	EN "helba" → AR هلبة may not match intended pronunciation or conventional rendering; flagged as transliteration error.
Semantics	Lexical Semantics	Unnatural / Unidiomatic style	Grammatically correct but sounds non-native/awkward; fluency degraded without direct grammar violation.	English phrasing "a girl from Saad El Dali" mis-conveys kinship; should be "Saad El Dali's daughter."
Semantics	Lexical Semantics	Awkward style	Overly wordy/embedded clauses; source-style retained inappropriately; redundancy.	Repetition "will be implemented, it will be implemented" mirrors Arabic emphasis but is stylistically awkward in English.

Semantics	Lexical Semantics	Coverage: Unknown term / dialect	Model fails to recognize a dialect-specific lemma; meaning distorted.	Egyptian خيابة rendered “audacity” (opposite valence).
Semantics	Lexical Semantics	Disambiguation: Polysemy failure	Word is known but wrong sense is selected in context.	Moroccan حارة taken as “warmth” (thermal) instead of intended “hard/harsh/intense treatment.”
Semantics	Lexical Semantics	Disambiguation: Cross-variety interference	Picks standard meaning instead of dialect meaning.	UAE باخذ إجازة used for “I will be off,” importing standard “formal leave/holiday” sense rather than dialect nuance.
Semantics	Lexical Semantics	Unintelligible	Output is garbled/ungrammatical/incomprehensible; meaning cannot be recovered.	Egyptian target includes invented/garbled form هيديكرم causing incomprehensibility.
Semantics	Lexical Semantics	Measurement units	Locale-inappropriate unit/format (e.g., feet/inches vs metric; decimal separator; % placement).	Examples include decimal separator comma vs period and % formatting differences.
Semantics	Propositional Semantics	Addition	Adds information not in source; changes truth-conditional content.	Mauritanian source → English adds “new moon’s arrival,” not present in source.
Semantics	Propositional Semantics	Omission	Drops source content; loss of propositional meaning (can remove critical info).	Palestinian example: translation loses proposition مكورن (“infected with corona”) and substitutes unrelated content.
Semantics	Propositional Semantics	Untranslated	Source-language segment carried over into target without translation.	English target leaves Arabic proper noun عبيد as-is in otherwise English sentence.
Semantics	Propositional Semantics	Hallucination	Fabricates new concrete details/props/actions not in source (often more specific than “addition”).	EN “If the issue isn’t worth it...” → Egyptian adds metaphor ميكسرش إزاة not present.
Semantics	Discourse Semantics	Inconsistent use of terminology	Uses multiple terms for same concept where consistency is required.	Example: “brake release lever” vs “brake disengagement lever” vs “manual brake release”...
Semantics	Discourse Semantics	Inconsistent with	Violates specified termbase/terminology guideline.	Example: termbase mandates “USB memory

		terminology resource		stick” but text uses “USB flash drive.”
Semantics	Discourse Semantics	Pronouns	Incorrect pronouns or reference tracking; changes speaker/referent/gender.	Moroccan ...واش انا عارفاني rendered “Am I not aware...” → shifts speaker–listener dynamics.
Semantics	Discourse Semantics	Inconsistent style	Style varies inconsistently across text (formal/informal; technical/colloquial; terse/wordy).	Example given: one part terse, others wordy; repeated segments translated differently across translators.
Morphosyntactic	Grammar (agreement/tense/etc.)	Verbal Features	Violates target grammar in verb system (tense/aspect/voice/mood/person; argument structure/valency).	EN “they reconciled” → Egyptian صالحوا without required object; should be اتصالحو or transitive with complement.
Morphosyntactic	Grammar (agreement/tense/etc.)	Nominal Features	Violates noun/NPh rules: number/gender/case/definiteness/state (idafa).	Mauritanian خياتي (fem. plural “my sisters”) translated “my brothers” → gender feature flipped.
Morphosyntactic	Constituent order	Address format	Locale-inappropriate address order/format; missing abbreviations; unconverted numerals; untranslated address parts.	Arabic address order retained in English (“Riyadh, Al Narjis, ...”) instead of English ordering; plus numeral/script issues.
Morphosyntactic	Constituent order	Date format	Uses wrong locale date convention (e.g., day/month/year vs month/day/year).	Arabic date ٢٢/١٠/٢٠٢٢ → U.S. English should be 10/23/2022 (not 23/10/2022).
Orthographic / Writing conventions	Spelling	Typo	Incorrect written word form by accepted target conventions.	Egyptian: شك → شكاك; فاء → هاء; خالده → في; corrected: شكاك في خالد...الدالي
Orthographic / Writing conventions	Spelling	Inconsistent spelling	Same word spelled multiple ways within same text (often proper nouns/loanwords/transliterations).	Defined in guidelines (no concrete MT sentence example provided there).
Orthographic / Writing conventions	Spelling	Unconventional spelling	Non-standard/idiosyncratic spelling when conventional form exists; harms readability.	Defined in guidelines (no concrete MT sentence example provided there).
Orthographic / Writing conventions	Surface mechanics	Number format	Locale-inappropriate number writing/formatting (digits,	Egyptian “two thousand and ninety-nine” → ألفين

			directionality, separators, grouping).	(incorrect) → وتسعة وتسعين ألفين تسعة وتسعين (correct).
Orthographic Writing conventions	Surface mechanics	Currency	Wrong currency rendering/format for locale (incl. misinterpreting a currency term).	EN “ten rubies” treated as gemstones rather than currency; should recognize monetary unit (e.g., روبيات).
Orthographic Writing conventions	Surface mechanics	Time format	Wrong time convention/formatting for locale (12/24h, zeros, AM/PM, ISO elements, numerals).	Examples: midnight labeled PM; missing leading zeros; leaving Arabic-Indic digits in English; missing timezone delimiter.
Orthographic Writing conventions	Surface mechanics	Telephone	Telephone number formatted incorrectly for locale (country code, prefixes, digit script, grouping).	Saudi numbers need +966 and dropping leading “0”; avoid Arabic-Indic digits in English output.
Orthographic Writing conventions	Punctuation	—	Punctuation violates target conventions; harms readability/flow; avoid copying source punctuation blindly.	Egyptian: extra commas fragment sentence: مش...مناسب، قلتك، معايا قلتك comma after قلتك disrupts flow.
Graphetic	Character encoding	—	Characters garbled due to encoding/parsing issues; often mixes scripts in corrupted ways.	English target begins with “Fواز” (Latin F inserted into Arabic name).